İşlenilmiş Konuların Tekrarı Niteliğinde Sorular

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1. Reel sayıları nasıl inşa ederiz? (How do we construct the real numbers?)
2. **Cantor's Power Set Theorem**: $|A| < |\mathcal{P}(A)|$
   (İpucu: $S = x \in A \mid x \notin f(x)$. No surjection $f : A \to \mathcal{P}(A)$ can exist.)
3. Let A and B be non-empty bounded subsets of $\mathbb{R}$, and define $A + B = \{a + b \mid a \in A, b \in B\}$. Prove that $\sup(A + B) = \sup(A) + \sup(B)$.
4. All subsets of $\mathbb{Z}$ are either finite or countable.
5. **Cardinality Addition:** $\aleph_0 + \aleph_0 = \aleph_0$
6. **Cardinality Multiplication:** $\aleph_0 \cdot \aleph_0 = \aleph_0$
7. **Continuum Cardinality:** $2^{\aleph_0} = |\mathbb{R}|$

   Sequence Limits & Convergence
8. Using the definition of convergence of a sequence, prove that $\lim_{n\to\infty} \frac{n}{2n+1} = \frac{1}{2}$.
9. What is the definition of a sequence?
10. State the Monotone Convergence Theorem. Show how it can be used to prove that the sequence $a_n = \sum_{k=1}^{n} \frac{1}{k^2}$ converges, even without knowing its exact sum limit.
11. State the Bolzano-Weierstrass Theorem. Give an example of a bounded sequence in $\mathbb{R}$ that does not converge, but from which you can extract at least two different convergent subsequences.
12. Prove that $\mathbb{R} \sim \mathbb{R}^2$. (ie. $|\mathbb{R}| = |\mathbb{R}^2|$)

    Bonus Question: Transcendental sayı ne demektir? Varlığını ispatlayınız. (İpucu: Cebirsel sayılar sayılabilirdir gösteriniz.)